

Current Biology, Volume 27

Supplemental Information

**Exceptional Evolutionary Expansion
of Prefrontal Cortex in Great Apes and Humans**

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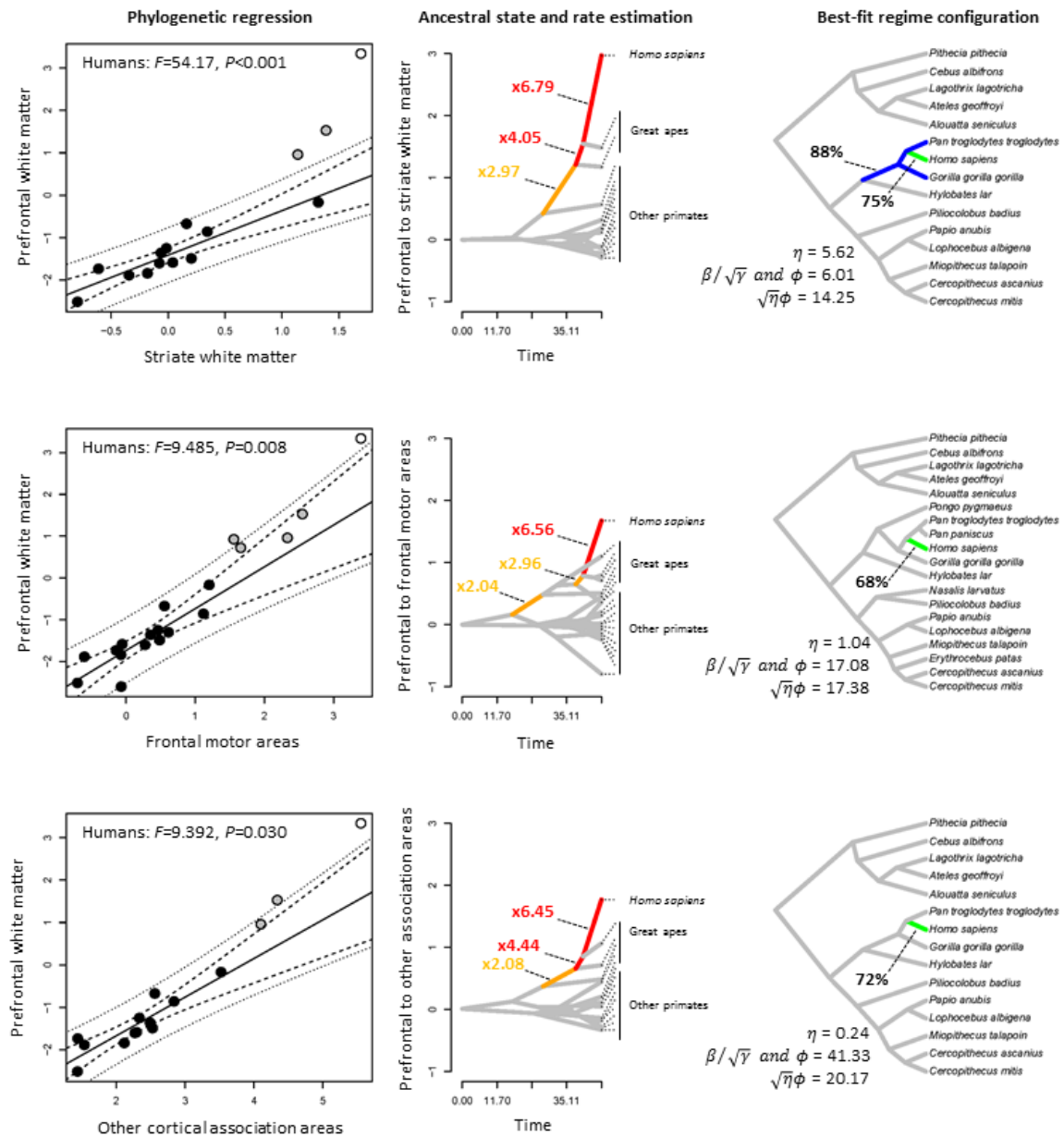


Figure S1: Evolutionary modelling of white matter. Related to Figures 2-3.

As in Figure 2, but for frontal white matter. Data from Smaers and colleagues [S1, S2].

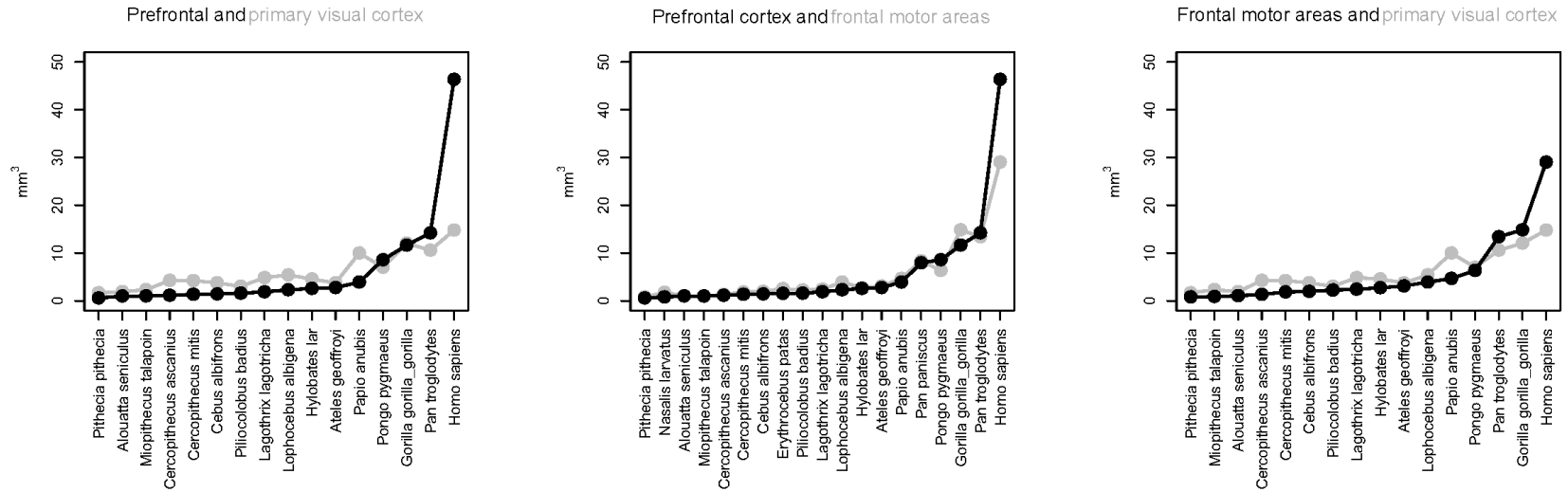


Figure S2: Expansion of prefrontal size in the human lineage. Related to Figures 2 and 4.

As in Figure 4, but for the Smaers data [S1, S2]. We note that, per design, the proxy for prefrontal cortex and frontal motor areas in this data are such that prefrontal cortex volume is underestimated, and frontal motor areas overestimated for great apes and humans. See first section on ‘Data’ in the supplementary information. These proxies thus underestimate trends in prefrontal cortex expansion.

Table S1: **Overview of data.** Related to Figure 1.

Volumes are in mm³, surface area in mm².

Smaers data	Gray matter volume				White matter volume			
	Primary visual	Prefrontal	Other cortical association areas	Frontal motor	Primary visual	Prefrontal	Other cortical association areas	Frontal motor
<i>Cercopithecus ascanius</i>	4.272	1.154	16.060	1.351	0.834	0.159	8.297	0.930
<i>Cercopithecus mitis</i>	4.232	1.380	19.385	1.852	1.042	0.205	9.911	0.946
<i>Erythrocebus patas</i>		1.560		2.552		0.272		1.848
<i>Miopithecus talapoin</i>	2.339	1.018	12.218	0.906	0.709	0.152	4.642	0.548
<i>Alouatta seniculus</i>	1.918	1.015	12.394	1.077	0.543	0.177	4.212	0.866
<i>Ateles geoffroyi</i>	3.753	2.773	32.933	3.119	0.985	0.286	10.332	1.586
<i>Lagothrix lagotricha</i>	4.856	1.887	24.188	2.434	1.226	0.226	12.496	1.625
<i>Cebus albifrons</i>	3.764	1.434	21.024	1.977	0.939	0.257	12.053	1.425
<i>Gorilla gorilla gorilla</i>	12.062	11.642	125.539	14.854	3.123	2.608	60.386	10.295
<i>Homo sapiens</i>	14.787	46.323	271.891	29.013	5.439	28.135	258.218	29.755
<i>Hylobates lar</i>	4.574	2.588	28.424	2.744	1.176	0.510	12.915	1.743
<i>Pan paniscus</i>		7.956		8.356		2.056		5.243
<i>Pan troglodytes troglodytes</i>	10.593	14.209	112.750	13.407	4.004	4.607	76.600	12.780
<i>Pongo pygmaeus</i>	7.008	8.596	76.924	6.353		2.520		4.745
<i>Pithecia pithecia</i>	1.679	0.600	9.242	0.819	0.451	0.081	4.200	0.496
<i>Lophocebus albigena</i>	5.420	2.311	29.102	3.926	1.411	0.424	17.092	3.063
<i>Papio anubis</i>	10.001	3.921	45.240	4.686	3.740	0.845	33.835	3.325
<i>Nasalis larvatus</i>		0.813		1.749		0.074		0.933
<i>Ptilocolobus badius</i>	3.053	1.591	22.629	2.260	0.926	0.201	9.665	1.317

Brodmann data	Surface area			
	Primary visual	Prefrontal	Other cortical association areas	Frontal motor
<i>Callithrix jacchus</i>	341	148	993	167
<i>Saimiri sciureus</i>	744			
<i>Cebus apella</i>	1786	1260	8814	1822
<i>Macaca mulatta</i>	1866	1733	9892	1817
<i>Papio cynocephalus</i>	2559	2111	13506	2200
<i>Papio hamadryas</i>	2130	1967	13599	2898
<i>Mandrillus sphinx</i>	3537	2168	23422	2194
<i>Hylobates lar</i>	1899	1839	10912	1651
<i>Pan troglodytes troglodytes</i>	3212	6719	24252	5389
<i>Homo sapiens</i>	3495	30234	68355	6117

Table S2: **Phylogenetic analysis of covariance.** Related to Figures 2-3.

Results from a phylogenetic analysis of covariance. Results relate to tests of differences in intercept among groups with the slope held constant. ‘Others’ refers to all non-great ape primates in the sample. The analysis includes the comparison of multiple treatment groups (group a ‘v’ group b) to a control group (‘|’ group c). Data from Smaers et al.[S1, S2] and Brodmann [S3].

		Pgls ancova							
		Smaers data				Brodmann data			
Y	X	Grouping	F	P		F	P		
Maximum likelihood estimation of λ									
Prefrontal	Primary visual	Among groups	15.630	<0.001	***	35.728	0.001	**	
		Humans v others great apes v others	30.622	<0.001	***	65.373	<0.001	***	
		Humans v great apes others	13.969	0.003	**	17.087	0.009	**	
		Great apes v others humans	14.431	0.003	**	10.453	0.023	*	
	Frontal motor	Among groups	4.842	0.024	*	24.911	0.003	**	
		Humans v others great apes v others	9.098	0.009	**	49.820	<0.001	***	
		Humans v great apes others	5.050	0.040	*	23.080	0.005	**	
		Great apes v others humans	5.994	0.027	*	2.316	0.189		
	Other association areas	Among groups	3.306	0.072		7.049	0.035	*	
		Humans v others great apes v others	6.389	0.027	*	13.431	0.015	*	
		Humans v great apes others	5.697	0.034	*	3.661	0.114		
		Great apes v others humans	1.931	0.190		6.162	0.056		
	$\lambda=0$								
	Prefrontal	Primary visual	Among groups	19.960	<0.001	***	35.728	0.001	**
			Humans v others great apes v others	32.142	<0.001	***	65.373	<0.001	***
			Humans v great apes others	10.653	0.007	**	17.087	0.009	**
Great apes v others humans			22.342	<0.001	***	10.453	0.023	*	
Frontal motor		Among groups	4.842	0.024	*	24.911	0.003	**	
		Humans v others great apes v others	9.098	0.009	**	49.820	<0.001	***	
		Humans v great apes others	5.051	0.040	*	23.080	0.005	**	
		Great apes v others humans	5.994	0.027	*	2.316	0.189		
Other association areas		Among groups	3.306	0.072		10.395	0.017	*	
		Humans v others great apes v others	6.389	0.027	*	16.525	0.010	*	
		Humans v great apes others	5.697	0.034	*	2.785	0.156		
		Great apes v others humans	1.931	0.190		8.273	0.035	*	
$\lambda=1$									
Prefrontal		Primary visual	Among groups	16.143	<0.001	***	28.728	0.002	**
			Humans v others great apes v others	28.776	<0.001	***	40.753	0.001	**
			Humans v great apes others	23.329	<0.001	***	39.433	0.002	**
	Great apes v others humans		9.180	0.011	**	5.305	0.070		
	Frontal motor	Among groups	4.569	0.028	*	17.291	0.006	**	
		Humans v others great apes v others	7.297	0.016	*	18.145	0.008	**	
		Humans v great apes others	8.438	0.011	*	28.936	0.003	**	
		Great apes v others humans	1.908	0.187		0.989	0.366		
	Other association areas	Among groups	4.417	0.037	*	5.484	0.055		
		Humans v others great apes v others	5.545	0.036	*	10.574	0.023	*	
		Humans v great apes others	8.827	0.012	*	6.210	0.055		
		Great apes v others humans	1.122	0.310		4.236	0.095		

Table S3: **Phylogenetic analysis of covariance**. Related to Figures 1-3.

As in Table S2 considering a maximum likelihood estimation of λ , but for the cumulative prefrontal volume of the 4th and 6th interval along the frontal pole.

<i>Smaers (2011) data, 4th interval</i>			Pgls ancova					
Y	X	Grouping	Gray matter			White matter		
			<i>F</i>		<i>P</i>	<i>F</i>		<i>P</i>
Prefrontal	Primary visual	Among groups	16.354	<0.001	***	25.640	<0.001	***
		Humans v others great apes v others	32.311	<0.001	***	49.399	<0.001	***
		Humans v great apes others	15.526	0.002	**	20.038	0.001	**
		Great apes v others humans	14.292	0.003	**	18.456	0.002	**
	Frontal motor	Among groups	6.371	0.011	*	4.867	0.025	*
		Humans v others great apes v others	11.270	0.005	**	8.808	0.010	*
		Humans v great apes others	5.637	0.032	*	4.916	0.044	*
		Great apes v others humans	8.926	0.010	*	6.724	0.021	*
	Other association areas	Among groups	5.607	0.021	*	5.023	0.031	*
		Humans v others great apes v others	10.977	0.007	**	9.714	0.011	*
		Humans v great apes others	9.347	0.011	*	4.870	0.052	
		Great apes v others humans	3.678	0.081		5.968	0.035	*

<i>Smaers (2011) data, 6th interval</i>			Pgls ancova					
Y	X	Grouping	Gray matter			White matter		
			<i>F</i>		<i>P</i>	<i>F</i>		<i>P</i>
Prefrontal	Primary visual	Among groups	2.925	0.096		28.177	<0.001	***
		Humans v others great apes v others	5.836	0.034	*	54.300	<0.001	***
		Humans v great apes others	3.902	0.074		22.051	<0.001	***
		Great apes v others humans	3.036	0.109		20.252	<0.001	***
	Frontal motor	Among groups	5.054	0.022	*	4.113	0.039	*
		Humans v others great apes v others	9.033	0.009	**	7.549	0.016	**
		Humans v great apes others	4.383	0.055		4.240	0.059	
		Great apes v others humans	6.997	0.019	*	5.862	0.030	*
	Other association areas	Among groups	2.925	0.096		5.074	0.030	*
		Humans v others great apes v others	5.836	0.034	*	9.724	0.011	*
		Humans v great apes others	3.902	0.074		4.682	0.056	
		Great apes v others humans	3.036	0.109		6.262	0.031	*

Table S4: **Testing for the effect of overfitting.** Related to Figures 2-3.

Results from a multi-regime OU model fitting analysis (using the approach proposed by Khabbazzian et al. [S42]) comparing the estimated model (presented in Figures 2 and 3) to an overfit model in which, in addition to the estimated regime configuration, New World monkeys were allocated to a separate regime. Results demonstrate that the overfit model consistently yields a lower pBIC score, confirming that the OU analyses adequately guard against overfitting.

Data	Y	X	pBIC estimated model	pBIC overfit model	Δ pBIC
Smaers data	Prefrontal	Primary visual	26.58	36.47	9.88
		Frontal motor	30.16	37.86	7.71
		Other cortical association areas	-3.49	5.07	8.56
Brodmann data	Prefrontal	Primary visual	19.5	26.12	6.62
		Frontal motor	12.36	18.30	5.93
		Other cortical association areas	15.4	21.49	6.10

Table S5: **Phylogenetic analysis of covariance for white matter.** Related to Figures 2-3.

As in Table S2 considering a maximum likelihood estimation of λ , but for white matter volume instead of gray matter volume. Data from Smaers et al.[S1, S2]

Pgls ancova				
X	Grouping	White matter		
		F	P	
Primary visual	Among groups	28.177	<0.001	***
	Humans v others great apes v others	54.299	<0.001	***
	Humans v great apes others	22.051	<0.001	***
	Great apes v others humans	20.252	<0.001	***
Frontal motor	Among groups	5.055	0.021	*
	Humans v others great apes v others	9.485	0.008	**
	Humans v great apes others	5.570	0.032	*
	Great apes v others humans	6.580	0.022	*
Other cortical association areas	Among groups	4.885	0.030	*
	Humans v others great apes v others	9.392	0.011	*
	Humans v great apes others	4.586	0.055	
	Great apes v others humans	5.952	0.033	*

Table S6: **PGLS regressions parameters**. Related to Figures 2-3.

Slopes and 95% confidence intervals of slopes derived from a phylogenetic generalized least-squares analysis of different subgroups in the sample.

Y	X	PglS						
		Sample	Gray matter			White matter		
			Slope	95% CI		Slope	95% CI	
Prefrontal	Primary visual	All	1.46	0.97	: 1.95	1.90	1.42	: 2.39
		Non-human	1.09	0.67	: 1.51	1.25	0.85	: 1.64
		Non-great apes	0.93	0.54	: 1.31	1.05	0.69	: 1.41
	Frontal motor	All	1.13	1.00	: 1.27	1.35	1.17	: 1.54
		Non-human	1.07	0.92	: 1.22	1.26	1.04	: 1.48
		Non-great apes	0.89	0.63	: 1.15	1.00	0.62	: 1.38
	Other association areas	All	1.18	1.11	: 1.26	1.28	1.12	: 1.43
		Non-human	1.14	1.06	: 1.23	1.18	0.98	: 1.38
		Non-great apes	1.07	0.92	: 1.22	0.94	0.63	: 1.25
	Rest of cortex	All	1.28	1.16	: 1.41	1.37	1.17	: 1.56
		Non-human	1.23	1.08	: 1.37	1.06	0.77	: 1.36
		Non-great apes	1.09	0.80	: 1.38	0.92	0.50	: 1.35
	Rest of brain	All	1.18	1.08	: 1.29	1.58	1.39	: 1.77
		Non-human	1.15	1.02	: 1.28	1.48	1.26	: 1.70
		Non-great apes	1.09	0.81	: 1.37	1.28	0.79	: 1.77

Supplemental Experimental Procedures

Data

Prefrontal and frontal motor data was taken from Smaers et al. [S1, S2] and Brodmann [S3] (Table S1). These data sets differ in the breadth of the comparative sample (19 versus 13 species, respectively), the nature of the measurement (mm^3 versus mm^2), and the cytoarchitectonic criteria that were employed (volumetric bootstrapping along the frontal pole [S2] relative to the cytoarchitectonic border between areas 3 and 4 [S1] versus granular and agranular cortex). Both data sets comprise one individual per species, except for the human data in Smaers [S1, S2] which comprises 8 individuals. Sex is unknown for the Brodmann data set. In the Smaers data, the nonhuman data comprise 7 females and 11 males and the human data comprise 3 females and 5 males. Although some work has suggested sex-by-hemisphere differences in the Smaers data set [S4], no significant sex differences with regards to whole prefrontal cortex volume have been observed.

Only 9 out of 13 species from the Brodmann data set can be used for the current purposes because in this data set not all measurements were taken for all species. The Smaers data set provides only a proxy of prefrontal cortex volume by employing volumetric bootstrapping along the frontal pole. This bootstrapping approach consists of considering the cumulative volume along a set of equidistant intervals along the frontal pole. This approach thus uses the border between area 3 (somatosensory cortex) and area 4 (primary motor cortex) as a functionally and neurobiologically relevant landmark relative to which prefrontal cortex (cumulative volumes posterior to anterior) and frontal motor areas (cumulative volumes posterior to anterior) are defined. This approach hereby accounts for the more caudal position (relative to monkeys) of the border between area 3 and 4 in the great ape and human brain. Any observed enlargement of prefrontal cortex in great apes and humans using this measure is conservative and likely to be an underestimation [S2] because, although validly accounting for the more caudal position of the border between area 3 and 4 in great apes and humans, it does not account for the more caudal position of the border between prefrontal and premotor cortex within the frontal lobe (thus providing an underestimation of prefrontal cortex volume and an overestimation of frontal motor areas in great apes and humans). Previous research using this approach has indicated that scaling patterns of prefrontal cortex volume are equivalent across the frontal pole [S5]. In the current analyses the 5th volumetric interval along the frontal pole was used (Table S2). Table S3 demonstrate that results are equivalent when using the 4th and the 6th volumetric interval.

Data on primary visual cortex was taken from de Sousa et al. [S6]. Other brain data was taken from Stephan et al. [S7]. All data in Smaers et al. [S1, S2], de Sousa et al. [S6] and Stephan et al. [S7] are derived from the same specimens housed at the C&O Vogt Institute for Brain Research [S8]. The phylogenetic tree was taken from Arnold et al. [S9].

Allometry

The extent to which one variable scales with another across a sample of species provides information as to how much one variable is expected to change given one unit of change in another variable. This is quantified by means of the slope of a regression analysis on log transformed data. Two variables are said to scale allometrically when their scaling coefficient is different from unity, and isometrically when it is equivalent to unity. The nature of the allometric relationship among brain structure sizes thus reveals the extent to which changes in one brain structure are expected to outpace the changes in another as overall brain size increases. The scaling coefficient hereby provides information on which changes in proportionality of brain structure sizes are expected given changes in overall brain size. Although traditional scaling studies mostly employed ordinary least squares regression for these purposes, more recent studies use phylogenetic generalized least squares ('pgls') regression to account for potential species dependence due to phylogenetic relatedness [S2, S5]. Specifically, pgls is a generalized least squares regression that uses the expected covariance due to shared phylogenetic history as the covariance matrix among observations in estimating the slope and intercept of a least squares regression [S10-S13].

Functionally and neurobiologically relevant measures of relative prefrontal cortex size

To evaluate whether a particular brain structure is enlarged relative to other structures, the standard approach has been to fit a regression line through a comparative sample and calculate to what extent predicted values correspond to observed values. This approach adequately accounts for scaling differences among different brain structures and is therefore an appropriate measure of relative size (i.e. independently of overall size). This approach was pioneered by

Passingham [S14] using standard least squares regression. Although this approach remains the primary means to assess relative size, more recent studies use phylogenetic generalized least-squares regression [S2, S5, S15].

Relative prefrontal cortex size has commonly been characterized as the size of prefrontal cortex relative to the size of the rest of the cerebral cortex or the rest of the brain [S15]. When aiming to investigate the extent of human prefrontal cortex expansion, this characterization of relative prefrontal cortex size does not, however, account for the functional and anatomical underpinnings of neural information processing [S5]. It has long been established that information processing in the brain occurs in a hierarchical manner. Stimuli are initially mapped onto primary sensory areas, and information subsequently projects to supplementary sensory areas and higher order association areas [S16]. Throughout this process stimuli become increasingly complex [S17] and more object-like [S18]. The increasingly complex information from the stimuli is subsequently integrated in higher order parietal and temporal association areas to form mental representations [S19]. Prefrontal cortex can be considered to stand at the top of this hierarchical mapping of information processing in that it controls the manipulation of, and changes in these mental representations, which, in turn, forms the basis of complex decision making [S20, S21]. The prefrontal cortex is therefore highly connected with other higher order cortical association areas such as parietal and temporal areas [S22, S23]. Therefore, comparing prefrontal cortex to the rest of the cerebral cortex (or rest of the brain) confounds the interpretation of whether prefrontal cortex is enlarged because a comparison is made with a measure that includes both areas with which it is highly integrated and areas with which it is not. A more appropriate comparison embedded within the functional and anatomical framework of neural information processing would be to compare target areas with their input [S5]. In the context of the prefrontal cortex, a comparison with primary sensory areas would reveal whether prefrontal cortex is enlarged relative to its sensory input (effectively comparing higher-order with lower-order information processing), and a comparison with other cortical association areas would reveal whether prefrontal cortex is especially enlarged relative to other heteromodal areas that project to it [S5]. Such results in our analyses unequivocally indicate that both prefrontal cortex and other cortical association areas are significantly expanded in great apes and humans relative to primary sensory areas, and that human prefrontal cortex is significantly expanded relative to other cortical association areas (Table S2, Figures 2-3).

To account for the pathway of information processing we consider four comparisons that together provide a comprehensive analysis of how prefrontal cortex volume has changed relative to other regions: (i) prefrontal cortex size relative to primary visual cortex; (ii) prefrontal cortex size relative to frontal motor areas; (iii) prefrontal cortex size relative to other cortical association areas; (iv) prefrontal cortex size relative to the rest of neocortex.

The comparison relative to the primary visual cortex, which represents the only primary sensory area for which comparative measurements are available, provides the most informative view of changes in prefrontal cortex size because it reliably reflects the hierarchical nature of neural information processing by comparing the amount of 'higher' associative processing capacities relative to the amount of 'lower' primary sensory information processing [S5] (see main text for more information). Similar to previous studies [S5] we consider the size of the primary visual cortex as indicative of the size of other primary sensory areas because it is a measure that is readily available across the primate sample and both have been shown to scale equivalently to body size [S14, S24]. Ideally, all primary sensory areas would be included in this measure, but this information is only available for 5 species of haplorhines [S25]. The fact that not all primary sensory areas are included in these analyses, however, results in a highly conservative test. Because all primary sensory areas are more closely aligned with changes in body size than brain size [S26], changes in visual primary sensory areas align more with other primary sensory areas than they do with cortical association areas.

The comparison of prefrontal cortex with frontal motor areas reflects the hierarchical information processing relationship between primary motor cortex and prefrontal cortex [S21]. However, no volumetric information is available separately on primary motor cortex. Our measure of frontal motor areas therefore includes elements of supplementary motor cortex and premotor cortex [S2, S27]. Because prefrontal and supplementary motor cortex are tightly interconnected [S28], this may confound the evaluation of the putative enlargement of prefrontal cortex volume relative to primary motor cortex. Any such confounding effect will, however, produce conservative results in terms of the expansion of prefrontal cortex relative to primary motor cortex such that any observed trend will be an underestimation.

The comparison of prefrontal cortex with other cortical association areas would reveal whether the prefrontal cortex has enlarged more than other cortical association areas with which it is tightly integrated (primarily higher order

parietal and temporal areas [S22, S23]). The proxy measure for ‘other cortical association areas’ that is used comprises neocortical size minus frontal lobe and primary visual size. Ideally, higher order parietal and temporal areas would be delineated using cytoarchitectonic or tract-tracing techniques, but standards to delineate these areas across species have yet to be developed.

The comparison with the rest of the cerebral cortex is susceptible to the same confounding effect as the comparison with frontal motor areas, but to a much higher degree. The ‘rest of the cortex’ includes both the higher order parietal and temporal areas with which the prefrontal cortex is highly integrated and the lower order primary sensory areas that form the basis of primary sensory information processing. Comparing prefrontal cortex to a measure that includes both higher order association areas and lower order primary sensory areas is functionally uninformative, problematic to interpret neurobiologically, and leads to an unrealistic level of conservatism in estimated patterns of change. More functionally and neurobiologically appropriate comparisons are with separate measures for other cortical association areas and primary sensory areas (as described above).

Phylogenetic analysis of covariance

Before interpreting allometry, formal tests are required to evaluate whether slopes and intercepts are homogenous in all subgroups of the sample [S29]. If such significant differences exist, the allometry may not reflect the most common relationship between size and form in the sample, but rather a departure of a particular subgroup. The consequence of not testing for homogeneity of intercepts and slopes would be that multi-grade isometry is mistaken for one-grade allometry. Indeed, comparing results of pgl regressions through different subsamples demonstrates how overlooking significant intercept differences among subgroups of the sample leads to the illusion that prefrontal cortex scaling is a one-grade allometry instead of a multi-grade isometry (Table S6).

Putative differences in slopes and intercepts among different groups in the sample can be tested using the method of general linear models in which analysis of variance is performed using least-squares regression [S30]. To test for differences in intercepts, we implement phylogenetic analysis of covariance by generalizing the standard generalized least squares procedure ($\mathbf{y} = \mathbf{X}\mathbf{b} + \boldsymbol{\epsilon}$) to include additional indicator variables describing group membership [S31]. In the context of comparing multiple treatment groups to a control group, the GLS regression model here becomes $\mathbf{y} = \mathbf{X}\mathbf{b} + \boldsymbol{\tau}_i + \boldsymbol{\epsilon}$ where $\boldsymbol{\tau}_i$ is the difference from group i to the reference group. $\boldsymbol{\epsilon}$ is hereby assumed to follow a multivariate normal distribution with mean of $\mathbf{0}$ (a null vector of length n) and an $n \times n$ phylogenetic variance-covariance matrix giving the variances and covariances among the observations. Results presented in Table S2 include the λ parameter in order to account for the degree of phylogenetic signal in the data [S12]. Table S2, however, demonstrates that results do not depend on λ (equivalent effects are observed when $\lambda=0$ and $\lambda=1$). Technical details and examples of implementation are available in Smaers & Rohlf [S31].

Phylogenetic analysis of covariance (‘ancova’) calculates the change associated with the clades of interest in the residual variance simultaneously with the phylogenetic regression parameters, thereby accounting for all available variance. This approach allows for a direct test of whether a model with multiple grades (assuming multiple groups with different mean trait values) provides a significantly better fit to the data than a model with only a single grade (assuming that no particular group indicates a significantly different mean trait value). By testing for differences among intercepts, phylogenetic ancova tests for the occurrence of grade shifts in phylogenetic data. It is important to emphasize that the current implementation of phylogenetic ancova uses least-squares procedures only [S31]. Least-squares phylogenetic ancova is hereby no different from standard least-squares ancova with the only exception that the phylogenetic variance-covariance is included in the error term. The advantage of using least-squares procedures is that it produces unbiased results irrespective of sample size. In the least-squares framework low sample size is not a statistical issue because least-squares minimizes the sum of squares of the residuals regardless of sample size and thereby provides unbiased estimates of the regression parameters. Because our implementation of phylogenetic ancova uses least-squares procedures only, this also applies to our analyses. Small sample size may of course, present empirical issues, though this is not the case for our analyses.

Least-squares phylogenetic ancova is preferred over methods that detect ‘evolutionary singularities’ [S32, S33] because such methods do not provide a direct test for differences in intercepts and slopes in a regression, nor do they provide a direct test comparing the fit of multiple grades relative to a single grade. Phylogenetic ancova is also preferred over the previously used procedure to remove the datum of interest, fit a regression line to the remaining

data points, and calculate prediction intervals for a new observation [S15, S34]. Such a procedure is not a correct statistical test because it does not account for the variance of the predicted means [S31].

Furthermore, because the procedure for phylogenetic ancova used here employs standard least-squares procedures only [S31], parameterization is automatically penalized within the estimation of the probabilities. Based on personal communications some have suggested that at least 10 data points are needed per parameter to be estimated [S35]. However, this cut-off is trivial and is suggested to apply to multidimensional data. Considering that our data are bivariate only, and that calculations are done using standard least-squares procedures only (which automatically penalize parameterization in the estimation of the probabilities), the parameterization of two or three grades is acceptable and poses no statistical issue.

Given the small sample size of the Brodmann data it may be purported that least-squares procedures do not adequately penalize for the added parameters assumed by including multiple groups in analyses. This question relates to the notion of whether degrees of freedom penalize for parameters in the least-squares framework in such a way to avoid overfitting. Considering that least-squares analysis is the golden standard in regression analysis, there is overwhelming support to accept this notion. We put this to the test for our data by testing whether a 3-grade model (human v great ape v other primates) does indeed provide a better fit than a 2-grade model (human v nonhuman), and whether a 4-grade model (human v great ape v other Old World primates v New World primates) provides a better fit than a 3-grade model. Note that an F-ratio test only tests whether a more parameterized model provides a significantly *better* fit than a less parameterized model (not vice versa). If the procedure tends to overfit, then a more parameterized model would always provide a significantly higher fit than a less parameterized model because the addition of more parameters would not be adequately penalized.

We tested this on the prefrontal-primary visual data. For both the Brodmann and Smaers data, we found that a 3-grade model does indeed provide a better fit than a 2-grade model (Brodmann: $F=10.4532$, $P=0.0231$; Smaers: $F=14.4314$, $P=0.0025$). This confirms our conclusion that both great apes and humans can be considered separate non-allometrically derived grades. Importantly, a 4-grade model is not significantly different from a 3-grade model (Brodmann: $F=0.9569$, $P=0.3834$; Smaers: $F\sim 0$, $P=0.9785$), demonstrating that least-squares phylogenetic ancova avoids overfitting in our analyses.

An assumption of the use of phylogenetic ancova to test for differences in intercepts among subgroups is that slopes are homogenous. The difference in the slope of the regression lines between the non-great ape sample and the great ape sample (including humans) is not larger than expected by chance for the Smaers data (Prefrontal vs primary visual: $F=2.712$, $P=0.126$; Prefrontal vs frontal motor: $F=0.911$, $P=0.355$; Prefrontal vs other cortical association areas: $F=2.263$, $P=0.158$). The Brodmann data set does indicate such differences (Prefrontal vs primary visual: $F=17.087$, $P=0.009$; Prefrontal vs frontal motor: $F=23.080$, $P=0.005$; Prefrontal vs other cortical association areas: $F=3.661$, $P=0.114$). The great ape sample in the Brodmann data set, however, consists of only 2 species (chimpanzee and human). Considering that scaling relationships among different variables and subgroups are equivalent in the Smaers and Brodmann data sets [S5] it is reasonable to assume that the difference in slopes in the Brodmann data set are due to $n=2$ in the great ape and human sample rather than representative of the true difference in slope (for which the Smaers data set gives a better estimate considering its larger comparative sample). Moreover, when a difference in slope is observed among subgroups, it can already be concluded that the subgroups belong to a different allometry (in such a case there is no need for a test of differences in intercepts to conclude that subgroups are part of a different grade). The interpretation of a phylogenetic ancova test for the Brodmann data set is therefore valid and provides a conservative estimate.

Evolutionary modelling

To provide additional tests for the occurrence of grade shifts we performed a series of different evolutionary modelling analyses. Using a phylogenetic tree and observed information from contemporary tip taxa, these evolutionary modelling approaches employ statistical and mathematical models of evolution to describe the pattern and rate of trait change along individual branches of a phylogeny. Specifically, we use Ornstein-Uhlenbeck ('OU') model fitting, and ancestral state and rate estimation methods. Each of these methods assume different models of evolution. OU model fitting assumes a multi-regime OU model. Ancestral state and rate estimation methods were used assuming two different multi-rate models (a multiple variance Brownian motion and a reversible-jump BM model), both of which produced equivalent results. These different methods add to our analyses by elucidating when in evolutionary time

deviations from allometric conservation occurred, and by demonstrating that the same results are obtained using different methods and different assumed models of evolution.

In all evolutionary modelling analyses, we calculated the relative size of prefrontal cortex for great apes and humans as the difference between the expected values based on the non-great ape sample relative to their observed values. We model the evolutionary expansion of prefrontal cortex volume in this manner for three reasons: (1) In order to investigate whether human prefrontal cortex follows general primate brain allometry, previous work [S15, S34] excluded the human value from the analyses and compared the predicted human value with the observed human value by assessing whether the latter lies within the confidence intervals of the former. In other words, the human value was assessed relative to the non-human allometry. Although the idea to exclude the human value of interest to assess whether the non-human data aligns with non-human allometry is valid, the approach of evaluating whether the observed value lies within or outside a confidence/prediction interval constitutes an incorrect test because it does not take into account the expected variance of the predicted means [S31]. Our approach to consider great apes and humans relative to the non-great ape/human pattern hereby mirrors the approach of previous work, but rather than performing an incorrect test on these values, we model these values in a manner that accounts for all variance. (2) Phylogenetic analysis of covariance unequivocally demonstrates that humans and great apes do not follow the same allometry as other primates (Table S2). This demonstrates that a multi-grade model is strongly preferred over a one grade model and therefore that residuals should be taken using a multi-grade model, not a one grade model. (3) The nature of the phylogenetic relationship between non-great ape primates and great apes (great apes are embedded within the earlier radiation of monkeys, see Figure 2) indicates that whatever trend is prevalent across monkeys is the ancestral condition of great apes. In order to assess the evolutionary expansion of great apes and humans the most appropriate test is thus to model their expansion relative to the ancestral non-great ape sample [S5, S36].

Evolutionary modelling: OU model fitting

The most frequently used statistical model of evolution is standard Brownian motion ('BM'), which assumes that traits change at each unit of time with a mean change of zero and unknown and constant variance [S37-S39]. Within Brownian motion, the evolution of a continuous trait 'X' along a branch over time increment 't' is quantified as $dX(t) = \sigma dB(t)$, where ' σ ' constitutes the magnitude of undirected, stochastic evolution (' σ^2 ' is generally presented as the Brownian rate parameter) and ' $dB(t)$ ' is Gaussian white noise. It is paramount to consider that, at a fundamental level, BM quantifies the accumulation of *variance* over time. Although novel phylogenetic comparative methods continue using BM as a baseline model, they incorporate additional parameters to model changes in *mean* value over time. By incorporating additional parameters to quantify differences in mean value, these more complex models (Ornstein-Uhlenbeck models, or 'OU' models) quantify possible deviations from the standard gradual mode of evolution assumed by BM. Specifically, OU models extend the standard BM model to include a parameter that represents the mean trait value (θ) and the rate at which changes in mean values are observed (α). The OU model hereby assumes that the evolution of a continuous trait 'X' along a branch over time increment 't' is quantified as $dX(t) = \alpha[\theta - X(t)]dt + \sigma dB(t)$ [S40]. The inclusion of parameters that estimate mean trait value (θ) and the rate at which changes in mean values are observed (α) renders the OU model framework more appropriate than BM for modelling changes in the direction of trait evolution. This is true because the OU model adequately differentiates between changes in the mean (θ and α) and variance (σ^2) of a trait over time. This standard OU model has been modified into multiple-regime OU models that allow modelling trait evolution towards different 'regimes' that each display a different mean trait value [S41]. Several methods have been developed that use this approach to model trait diversification by estimating shifts in θ values along the branches of the phylogeny [e.g., S42, S43].

Although OU modelling approaches are commonly agreed to be very powerful for modelling trait diversification, recent research has pointed towards some challenges in traditional methods. Specifically, the theoretical properties of the maximum-likelihood estimators for OU parameters can result in non-uniqueness and inaccuracy, and traditional model selection criteria tend to favor overly complex scenarios [S44]. More recent approaches, however, overcome these difficulties by using a least-squares procedure and modifying model selection criteria so that they guard against overfitting [S42, S44]. Khabbazian et al. [S42], for example, developed a least-squares procedure to detect shifts on individual branches and a phylogenetic lasso procedure to regularize the shift configuration across the tree. This procedure ensures statistical selection consistency, small prediction error and uniqueness of estimators. To avoid overfitting, this procedure further employs a phylogenetic Bayesian Information Criterion ('pBIC') that better estimates model posterior probability than the traditional Akaike Information Criterion ('AIC') when errors are phylogenetically correlated. This procedure has been demonstrated to be highly conservative in that it provides high

precision (the detected shifts are true) but a low recall rate (some true shifts are missed) [S42]. We use Khabbazian et al.'s approach in our analysis.

Given the small sample size of the Brodmann data it may also be purported that the analysis would not adequately penalize for the added parameters assumed by including multiple groups. This assertion is counter to evidence that pBIC reliably guards against overfitting [S42], but may deserve special scrutiny for our particular data set. To test putative overfitting, we compared the 'estimated model' (displayed in Figures 2 and 3) against a more parameter-rich model in which, in addition to the estimated regime configuration, New World monkeys are allocated to a separate regime ('overfit model'). If our analyses were prone to overfitting, the more highly parameterized overfit model would produce a better statistical fit to the data than the estimated model. We find, however, that, for each analysis for both the Smaers and Brodmann data, the estimated model provides a better fit than the overfit model ($\Delta\text{pBIC} > 5$; see Table S4). These results demonstrate that also our OU analyses are robust against overfitting.

Evolutionary modelling: Ancestral state and rate estimation

To visualize trait change over time we use multi-rate Brownian motion models to estimate values at internal nodes of the phylogenetic tree and rates of evolution of particular traits across the entire sample. Traditionally, standard BM models have been used for these purposes [S45]. A consequence of assuming standard BM is that average trait change is proportional to the square root of time and that the rate of change is constant such that it has a single mean and variance across all branches. There is, however, evidence to suggest that the direction and rate of some traits may change among different lineages [S46]. In such instances, the standard BM rate parameter is not expected to capture all meaningful evolutionary trait variation. To account for lineage-specific variation in direction and rate of evolution, recent methods have expanded the standard BM model to include the estimation of multiple rates along the phylogeny. Here we compare results of the reversible-jump MCMC procedure ('rjBM') introduced by Venditti et al [S47, S48], and the multiple variance BM ('mvBM') procedure introduced by Smaers et al. [S49]. Because estimated rates in the rjBM procedure can be interpreted as traditional standard BM rate parameters (σ^2), the multi-rate rjBM model can be parameterized using a standard BM model to estimate ancestral states. More information on this procedure and its implementation is available from the manual of the software package that supports this procedure [S48]. The MCMC procedure uses 10 million iterations, sampling every 100th iteration, ensuring normal distribution of log likelihood values.

Using the rjBM and mvBM procedures we further calculated the extent to which the rjBM and mvBM rate estimates for individual lineages differ from a baseline expectation of gradual evolution. The amount of change observed at each branch (calculated as the difference between ancestral values as inferred from mvBM or rjBM approaches and descendent values) was hereby compared with a neutral scenario in which all the species in the phylogeny were simulated to evolve at a constant rate [S50]. The original phylogeny was rescaled to generations by dividing each branch length (in Myrs) by the generation time corresponding to their descendent species or node. For internal nodes, ancestral generation times were calculated using a constant variance BM approach. Age at first reproduction obtained from van Schaik & Isler [S51], and missing values from the PanTHERIA database [S52], was used as a proxy for generation time because this parameter is available for all the species in our dataset. Once the phylogeny was rescaled in this way, a per-generation evolutionary rate (σ^2) was calculated based on each dataset. This constant rate was used to simulate evolutionary change across the phylogeny as a random walk with branching points at corresponding nodes. Simulations were run 1,000 times and the mean change between all ancestors and descendants was used as the expectation of the amount of change had each branch evolved neutrally under a pure BM model. rjBM- and mvBM-based measures of change per branch were compared with mean simulated amounts of change per branch by calculating a ratio between the former and the latter. A ratio greater than 1 indicates that a species has experimented a greater amount of evolutionary change than expected in a constant-rate neutral scenario regardless of the directionality of change, whereas a ratio smaller than 1 indicates that a species has changed less than expected. For the present data set, rjBM and mvBM approaches yielded similar results. Analyses were carried out in Mathematica 10 using the package 'Phylogenetics' [S53].

Power, effect size, and sample size.

Given the small sample size in the Brodmann data, it is important to clarify how such low sample size relates to the power, effect size, and thus reliability of our results.

The power of a test determines the probability of rejecting the null hypothesis when it is in fact false. In general, the issue of power only arises when results are *not* significant. The power of a test therefore helps determine the sample size that is needed in future experiments to reliably discriminate between the null hypothesis and the alternative hypothesis. The calculation of power thus relates to the research design of future experiments. One can also calculate the power after an experiment has already been concluded. In this case, the power is referred to as the *observed power*. This case relates to the analyses we present. But here also, the issue of observed power only arises when results are not significant. When the observed power is high in a non-significant test, this is sometimes used to argue in support of the null hypothesis. This interpretation is, however, inappropriate and is referred to as the “power approach paradox” [S54]. When results are, however, significant (as is the case in our results), the computation of observed power does not add information over the *P*-value because the observed power is a simple function of the observed *P*-value [S54]. In other words, when the *P*-value indicates that the test is significant, the observed power of the test is, per definition, high. Considering that our ancova results are significant, the observed power of our tests must be high.

Furthermore, as mentioned, the phylogenetic ancova procedure we employ uses least-squares procedures only. Least-square procedures minimize the sums of squares of the residuals and hereby always provide unbiased estimates of the regression parameters regardless of sample size. The fact that low sample size is not a statistical issue for least-squares analysis should not, however, be confounded with the uncertainty that low sample size causes for the maximum likelihood estimation of parameters of models of evolution. The estimation of the λ parameter, for example, has been shown to have adequate power only for trees larger than 20 tips [S55]. The inclusion of the λ parameter in phylogenetic least-squares analysis thus introduces an element of uncertainty. We show that for our analysis, results are not dependent on the λ parameter (equivalent results are obtained in both extreme cases $\lambda=0$ and $\lambda=1$, see Table S2). Because our results are not dependent on the λ parameter, they are also not dependent on the uncertainty due to low sample size in estimating the λ parameter. It is paramount that the uncertainty in the maximum likelihood estimation of λ due to small sample size is not confounded with the purported uncertainty in the least-squares calculation of the regression parameters. As mentioned, least-squares procedures provide unbiased calculations of regression parameters irrespective of sample size.

With respect to uncertainty due to low sample size in the estimation of parameters of models of evolution it is further crucial to adequately differentiate between different models of evolution. Different models make different assumptions about the evolutionary process, and thereby include different statistical parameters that represent those assumptions. As mentioned, the standard BM model of evolution assumes the evolution of a continuous trait ‘*X*’ along a branch over time increment ‘*t*’ is quantified as $dX(t) = \sigma dB(t)$. Considering that σ represents the dispersion of the normal distribution from which changes per unit of time are sampled, it is evident that a low sample size will hamper its reliable estimation. In other words, low sample size is expected to limit the reliable estimation of the one parameter that is assumed by a standard BM model in this regard (i.e. σ).

The case of the OU model (used in our analysis of best-fit regime configurations) should, however, be differentiated from the case of the standard BM model. As mentioned, the standard BM model quantifies *variance* (σ^2) over time whereas the OU model includes additional parameters (θ and α) to quantify changes in *mean* values over time. Crucially, the inclusion of these two extra parameters also influences the evaluation of power. Within an OU model, power is not dependent only on σ (as in BM), but also on θ and α . In this regard, several authors have demonstrated that power within an OU modelling framework is, in fact, better predicted by effect size than by sample size [S44, S56]. Several measures of effect size have been proposed. Ho & Ané [S57] proposed $\beta/\sqrt{\gamma} > 2$ as indicating high effect size. β hereby indicates the difference in mean trait value among regimes (or $\Delta\theta$), and γ represents the stationary variance ($\sigma^2/(2\alpha)$). Ho & Ané [S57] found that their measure of effect size is a better predictor of power than sample size. Similarly, Cressler et al. [S56] found that combining a measure of selection opportunity $\eta = \alpha T$ (where *T* is the total depth of the phylogeny) with an equivalent measure of effect size $\phi = \sqrt{2\eta}/\gamma = \sqrt{2\alpha}\Delta\theta/\sigma$ into a ‘signal-to-noise’ ratio ($\sqrt{\eta}\phi$) provided a better prediction of power than sample size (high power can be inferred when $\sqrt{\eta}\phi \gg 1$). Because OU modelling approaches primarily aim to detect shifts in *mean* values over time, OU modelling power thus needs to be addressed primarily in terms of effect size. If effect size is high (difference in mean trait values among groups is high), the power to discriminate among different groups is high. Within the OU model framework, power thus crucially depends on the magnitude of the difference in mean trait values among groups. In all our analyses, results demonstrate that effect size is exceptionally high. In other words, differences between humans, great apes, and other primates are so pronounced that alternative regime configurations have a low statistical fit.

Effect size can be put to the test using bootstrap analysis. Bootstrap analysis includes fewer assumptions than estimating regime configurations directly from the data, and can thus be considered as a conservative approach. Khabbazian et al [S42] propose a bootstrap analysis for OU modelling that consists of randomizing residuals across the tree, estimating shift configurations for each iteration, and calculating the proportion of bootstrap iterations when a shift was detected on a particular branch. Results are presented in Figures 2-3 and Figure S1. Bootstrap results further confirm the exceptionally high effect size observed in our data.

An additional way in which effect size can be estimated is to test the same hypothesis using different methods with different model assumptions. If the same result is obtained across different methods and model assumptions, it can be concluded that the effect size is high regardless of the methods used and models assumed. We followed this principle in our research design by testing the hypothesis of significant differences among humans, great apes and other primates in prefrontal expansion using four different methods and five different model assumptions. The four methods include phylogenetic ancova, ancestral estimation, rate estimation, and OU model fitting. The five model assumptions include standard BM [S10] and λ -adjusted BM [S12] (phylogenetic ancova), reversible-jump BM [S47] and multiple variance BM [S49] (ancestral and rate estimation), and multi-regime OU [S42] (OU model fitting). All approaches converge on the same conclusion that humans, great apes, and other primates are distinct grades in prefrontal expansion.

In general, although there is no statistical reason to doubt the reliability and validity of our results, it is clear that a larger sample size would further increase statistical confidence in the results of the analyses. We hope that the present results inspire future studies to continue to expand comparative neuroanatomical data sets.

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